



Animation tsunami modelling in 3D to determine shelter and evacuation routes

Muhammad Yudhi Rezaldi, Ambar Yoganingrum, Nuraini Rahma Hanifa, Abdurrahman Prasetyadi, Widjo Kongko & Eko Yulianto

To cite this article: Muhammad Yudhi Rezaldi, Ambar Yoganingrum, Nuraini Rahma Hanifa, Abdurrahman Prasetyadi, Widjo Kongko & Eko Yulianto (2025) Animation tsunami modelling in 3D to determine shelter and evacuation routes, *Environmental Hazards*, 24:4, 391-412, DOI: [10.1080/17477891.2024.2400140](https://doi.org/10.1080/17477891.2024.2400140)

To link to this article: <https://doi.org/10.1080/17477891.2024.2400140>



Published online: 24 Sep 2024.



Submit your article to this journal [↗](#)



Article views: 241



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)



Animation tsunami modelling in 3D to determine shelter and evacuation routes

Muhammad Yudhi Rezaldi^{a,b}, Ambar Yoganingrum^a, Nuraini Rahma Hanifa^c,
Abdurrahman Prasetyadi^a, Widjo Kongko^d and Eko Yulianto^c

^aResearch Centre for Data and Information Sciences, Electronics and Informatics Research Organisation, National Research and Innovation Agency, Bandung, Indonesia; ^bVisual Communication Design, Faculty of Creative Industries, Telkom University, Bandung, Indonesia; ^cResearch Centre for Geological Disaster, Earth and Maritime Research Organisation, National Research and Innovation Agency, Bandung, Indonesia; ^dResearch Center for Hydrodynamics Technology, Research Organisation of Energy and Manufacture, National Research and Innovation Agency, Yogyakarta, Indonesia

ABSTRACT

Indonesia is located on the Pacific Ring of Fire, making the country prone to earthquakes and tsunamis. Indonesia, especially the southern region of Java, has the potential to be hit by a megathrust earthquake triggering waves much higher than before. Determining evacuation routes and locations as a part of disaster management is essential to avoid many fatalities. This research aims to determine evacuation routes and shelters using photogrammetry-based 3D animation. This paper applied experimental research for modelling and simulation approach with the stages of generating a 3D orthomosaic map, predicting tsunami magnitude, determining the height and distribution of waves, modelling and simulating 3D animation as well as marking evacuation routes and shelter locations. 3D orthomosaics are created based on aerial photos (UAV) drones. Future tsunami data are used to estimate the extent and depth of inundation that may occur. This research uses the Teluk Penyu area in Cilacap, Indonesia, as a study site. The tsunami disaster caused by an earthquake magnitude 7.8 centred in the south of West Java hit this city in 2006. This research produces a 3D animation model that can visualise the extent and depth of tsunami inundation to determine evacuation routes and shelter buildings.

ARTICLE HISTORY

Received 28 May 2024
Accepted 29 August 2024

KEYWORDS

UAV; 3D orthomosaic;
evacuation route; shelters;
tsunami

1. Introduction

A 2006 tsunami earthquake magnitude 7.8 centred on the southern coast of West Java, Indonesia hit the *Cilacap* region producing waves of up to 5–7 metres and claimed many victims (Fritz et al., 2007). During this event, the wave height on the south coast of *Nusa Kambangan* Island, *Cilacap*, facing the Indian Ocean, reached a height of 21 metres (Fritz et al., 2007; Hébert et al., 2012). It is reported that in the future, the southern region of Java will experience a potential megathrust earthquake of 8.8 Mw (Widiyantoro et al., 2020). This earthquake will trigger a much higher potential for tsunami waves.

CONTACT Ambar Yoganingrum ✉ amba002@brin.go.id 📧 National Research and Innovation Agency, KST Samaun Samadikun, Jl. Sangkuriang, Bandung, Indonesia

© 2024 National Research and Innovation Agency

Therefore, to prevent many casualties, it is necessary to build evacuation routes and determine evacuation locations in the area. The local government has determined evacuation routes and shelters around the Teluk Penyu tourist area, Cilacap city. The appropriate evacuation is vertical evacuation considering the flat morphology of the city and densely packed with buildings. This evacuation directed people to evacuate to nearby buildings that were within the reach of the tsunami (Salmon, 1984; Sun et al., 2022). Thus, determining buildings based on future tsunami data will provide an accurate and relevant model for dealing with future disasters.

A technique developed by Marfai et al. (2021) using Unmanned Aerial Vehicles (UAV) and future tsunami data produced up-to-date and accurate inundation prediction model. However, they provide only two-dimensional data, so shelter height and water depth are not visible yet. Three-dimensional data are required especially for urban contexts so existing buildings can be assessed for their suitability as shelters accurately. In addition, 3D modelling has advantages creating simple to complex visualisations fast for various natural disaster grid computing applications (Pajorová et al., 2008). Furthermore, 3D modelling produces a good accuracy for the tsunami inundation model (Vilarinho et al., 2016). Combining geo-visualisation techniques and multimedia design will contribute to accurate and realistic visualisation and improve the delivery of disaster information to people (Rezaldi et al., 2023a). Information presented in 3D multimedia form, for instance, animation makes it easier for the general public to understand (Yoganingrum, 2018). Another advantage, 3D simulation allows modifying autonomous system scenarios for disaster rescue operations in real-time, so that their configuration can adapt to mission objectives (Bertolino & Tanzi, 2020).

Modelling techniques using existing topographic maps have weaknesses in updating maps, especially in developing rapidly areas. This technique was developed for several developing tsunami-prone areas in Indonesia such as *Pelabuhan Ratu* (Indira et al., 2023), *Padang* (Ashar et al., 2018), *Lampung* (Febrina et al., 2020), *Kulon Progo* (Jati et al., 2023) and *Cilacap* (Laksono et al., 2022). In addition, the availability of high-resolution topographic data for tsunami hazard modelling is still lacking (Marfai et al., 2018). Moreover, Marfai et al. (2021) show that the inundation model provided by the government based on the Geometry of the Indonesian Topographic Map is still not accurate. Other modelling determined based on terrain conditions, not related to the water distribution and levels model in the event of a tsunami (Jati et al., 2023; Traya et al., 2022). Even if it is associated, it uses assumed data or past data (Nurwatik & Setiawan, 2023; Usman et al., 2017).

This paper creates photogrammetry-based 3D animation using data on future tsunami events to determine evacuation routes and shelters in *Cilacap*. The photogrammetry process uses a UAV (drone), and then, the aerial photos obtained are processed using Agisoft Metashape software to produce a 3D orthomosaic map. UAV have proven to be an invaluable asset in disaster management due to their ability to provide fast, high-quality data, low-cost technology and up-to-date information. Utilising future tsunami data will produce accurate models in dealing with future disasters. Meanwhile, 3D animation makes it easier for stakeholders to understand the information presented. This research discusses the use of two drones each with a dual and circular grid mission to create a 3D orthomosaic, shows the creation of an animated film about the magnitude of a future tsunami using aerial photographs taken with a UAV – a photogrammetric

technique, and presents the use of animated films for identify buildings for shelter and evacuation routes.

2. Literature review

2.1. Study area

Cilacap city is situated on the southern coast of Central Java, Indonesia, with a dense population. The topography of *Cilacap* is relatively flat and has the potential to be hit by tsunami waves from three directions. This city borders the Indian Ocean to the south and west, and rivers to the south and east have the potential to become entrance points for tsunami waves. *Cilacap* operates a number of industries such as oil refining, agricultural sectors, tourism, marine, wood manufacturing, cement, woven thread manufacturing, wheat flour processing, fish canning, electricity generation, and many more (Hidayat et al., 2021; Kerta et al., 2021; Kongko, 2012; Laksono et al., 2022). *Cilacap* runs a steam power plant (Laksono et al., 2022), an international standard sea port centre (Kongko, 2012), and a railway network. Moreover, a number of buildings such as hospitals, government offices and hotels are scattered just a few metres from the beach.

The research site, *Teluk Peny*, is one of the popular tourist destinations in the south of *Cilacap*, which borders the Indian Ocean (Figure 1). This site is a tsunami-prone area, which is busy with tourists at certain times. *Teluk Peny* offers visitors scenic beauty, recreational activities and local culture. This area is also a densely populated area, and there are several important public facilities such as hotels, markets, schools, shops, monastery and offices. The run-up height of the 2006 tsunami measuring 6.8 was 2.3 metres with insignificant inundation heights. However, the inundation will be wider and higher in

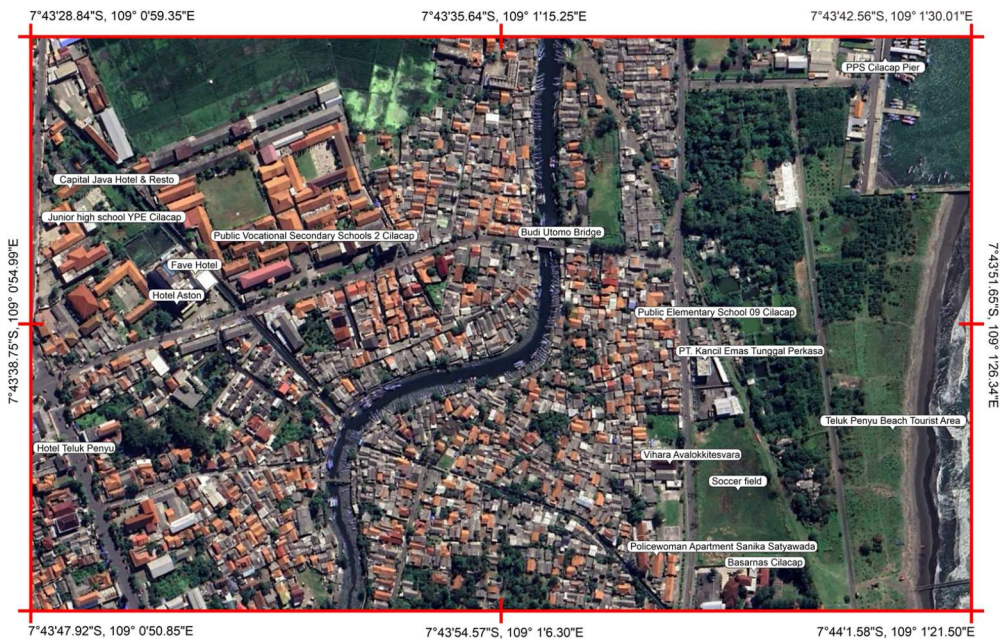


Figure 1. Research location area. Source: Authors analysis.

the future megathrust earthquake estimated to have a magnitude of 8.8 on the Richter scale (Widiyantoro et al., 2020). This area is likely to be attacked by tsunami from the south and river in the east.

2.2. UAV platform and digital camera

Unmanned Aerial Vehicles (UAV) also called Unmanned Aircraft Systems (UAS) or drones. Drones are categorised as robots that are capable of flying independently, controlled by software embedded in the system and equipped by high-resolution cameras, sensors, and Global Positioning System (GPS). They can provide real-time aerial imagery and mapping of an area; therefore, drones are a valuable tool for studying tsunami characteristics (Danardono et al., 2016; Marfai et al., 2018; Marfai et al., 2019a; Marfai et al., 2019b, 2021; Rezaldi et al., 2021). Kind of cameras used in previous research are Canon Powershot A2500 pocket camera with a resolution of 16 Mpix as the sensor (Marfai et al., 2018, 2019b, 2021) and then the DJI Phantom 4 and DJI Mavic Pro, Camera Sensor 1/2.3 (CMOS), Adjustable Aperture $f/2.8$ and $f/2.2$ (Rezaldi et al., 2021).

There are two important factors considered to produce accurate and quality aerial photos for 3D orthomosaic, namely shooting techniques and ground control point (GCP) (Elhalawani et al., 2021; Nogueira et al., 2017). The appropriate image capturing technique for buildings and tree environments is close-range photogrammetry (CRP) and camera orientation measurement by concern to the specifications of the camera used (Elhalawani et al., 2021). For example, a Sony Cybershot HX7 V digital camera that has been calibrated first must pay attention to the baseline/distance ratio – the distance between the two camera positions when taking the photo until the distance between the camera and the object comparison is within reasonable limits. The correct ratio value is between 1/15 and 1/20. For CRP projects, photos must be taken according to the '3X3 Rule' (Gurjar et al., 2013). Meanwhile, the Ground Control Point (GCP) is a tool for receiving signals in the form of sine waves from GPS satellites and processing them into data. This tool produces images with a high degree of accuracy when using multiple GCP tie points. These tie points create a mosaic in aerial photos (Júnior et al., 2018; Nogueira et al., 2017). Five GCP with a specification area of 3.85 ha using the DJI Phantom 4 Advanced UAV, flying height of 31 m, lateral and longitudinal overlap of 90% and 80% respectively, and using 22 quality checkpoints, resulting in an accuracy of 0.023 m for X, 0.031 m for Y, and 0.033 m for Z (Garcia & Oliveira, 2020). Meanwhile, the GCP pattern, which is distributed evenly throughout the field, provides the best calibration on 18 ha of agricultural land (Zhang et al., 2022).

2.3. Photogrammetry, orthophotos and orthomosaic

The American Society for Photogrammetry and Remote Sensing defines Photogrammetry as the combination of art, science and technology to obtain information about physical objects and the environment through the process of recording, measuring and interpreting photographic images, such as digital images, radiated acoustic energy patterns, laser range measurements and recorded emission patterns of electromagnetic energy, including magnetic phenomena (Wolf et al., 2014). Captain Eduard Deville, Surveyor General of Canada, discovered photogrammetry for topographic

mapping in 1886. This technique developed rapidly and was mass produced in the period between World War I and World War II. Photogrammetry is the most accurate and efficient method for compiling maps and producing topographic information (Wolf et al., 2014).

Photogrammetric processing using the Digital Terrain Model (DTM) produces orthophotos, while using the Digital Surface Model (DSM) produces true orthophotos. The difference is that orthophotos provide information such as the slope of the building and the height of the structure. Meanwhile, true orthophoto captures a vertical view of almost any point on the Earth's surface, mapping every shape and object visible on the ground (including vegetation and artificial objects). In the context of photogrammetry using drones, it is usually called orthomosaic, similar to true orthophoto (because it is produced using a DSM). Orthomosaics map can be generated based on nadir (vertical) and tilted images (oblique) (Buczowski, 2022). The map consists of detailed and georeferenced images, created by combining high-resolution photographs taken from drones or other aerial platforms. The creation process uses software and photogrammetry technique. There are 2 types of orthomosaic, namely 2D – a very sharp, highly detailed top-down image with precise coordinates – and 3D – like a 2D orthomosaic but with an extra dimension height.

2.4. Animation as an analytical tool

Animation as an analytical tool is something that has not been widely used in the disaster field. Animated films are used to convey information about hazards, evacuation, shelter position, etc. to improve preparedness people to face the disaster (Rezaldi et al., 2021; Shuto, 2019). Tsunami information presented in animated form is easy for adults and school children to understand (Affan et al., 2012; Goto et al., 2010; Goto et al., 2012). Animated films are used to increase elementary school students' preparedness for tsunami disasters (Amri et al., 2020; Idrus et al., 2019). Meanwhile, an animated exploration map was created to explain the interpolation method for distance measurements between forest fires to threatened communities (Kim et al., 2006). An initial study of orthomosaic map animation was carried out by Prasetyadi et al. (2022). This research has been hampered because it has not succeeded in creating a 3D orthomosaic map. Meanwhile, Kawai et al. (2016) reported that the Tsunami evacuation training system with animated simulations showed good results when tested on adult participants. In general, tsunami animated films are created to increase perceptions of efficacy such as tsunami warning signs, providing knowledge about evacuation routes and final shelter positions.

2.5. Shelters and routes in evacuation planning and management

There are two types of tsunami evacuation, namely horizontal and vertical areas. Horizontal means evacuate beyond the reach of the tsunami. Meanwhile, vertical is a safe areas still within the reach of the tsunami (Salmon, 1984; Sun et al., 2022). Several other considerations that must be taken into account in determining the evacuation location are the ability of humans to run long distances in panic conditions, there will be traffic jams if using vehicles, because at the same time everyone is going to the shelter location,

and the possibility of damage to roads and buildings due to the earthquake will make it difficult for people to reach refugee locations (Wood et al., 2014).

3. Methods

The experimental research method is used in this research through approach modelling and simulation that are known to be effective in designing and evaluating complex systems and processes (Menner, 1995). This method is commonly used to study natural disasters, especially tsunami (Marras & Mandli, 2021). The stages of the method in this paper are generating 3D orthomosaic map, predicting tsunami magnitude, determining height and distribution of waves, modelling and simulating 3D animation and then marking evacuation routes and shelters locations. Each step is approached systematically for successful modelling and simulation.

This study uses photogrammetry techniques to create a 3D orthomosaic map. Photogrammetry is a mapping technique using aerial photography. In this study, aerial photos were taken using a camera mounted on a UAV or drone. This type of photogrammetry measures photos precisely to determine the relative location of a point. This method determines the distance, angle, area, volume, height, and size and shape of an object (Wolf et al., 2014).

3.1. Equipment

The types and specifications of equipment used are as follows:

- This research used two types of drones, namely the DJI Phantom 4 Pro Obsidian and Mavic Pro. The DJI Phantom 4 Pro Obsidian takes aerial photos using the close-range photogrammetry (CRP) technique and a circular mission - a technique for taking photos at close range while surrounding an object. This drone has a 1-inch CMOS sensor with a resolution of 20 MP to record images, Gimbal-Stabilized 4K60/20MP Imaging - a sensor to stabilise images to prevent blurring, and visual object tracking - a sensor to detect object movements and changes occurring around the object. Apart from that, the drone has five redundant sensors to avoid collisions in the air. The drone can cover a maximum distance of 4.3 miles (7 km). Meanwhile, the battery capacity is up to 30 min. In the meantime, the Mavic Pro takes aerial photos with a double grid mission - capturing the object from a high angle. Figure 2 shows the aerial photos with a double grid and circular mission.
- We used a geodetic device - GPS Geodetic GNSS RTK Efix - F4 that receives signals in the form of sine waves from GPS satellites. GPS receivers are positioned on the earth's surface to receive signals and process them into data. This data is then used as the Ground Control Point (GCP) coordinates.
- The laser telescope type of Li-Ion, SND WAY, SW-600A (3-600 m) is a telescope for measuring the height and distance of buildings recommended as shelters. This tool can measure target distances between 3 and -600 m, with an accuracy of ± 1.0 m, an angle range of ± 60 degrees, a speed measurement accuracy of ± 5 km per hour and a magnification of 7X.

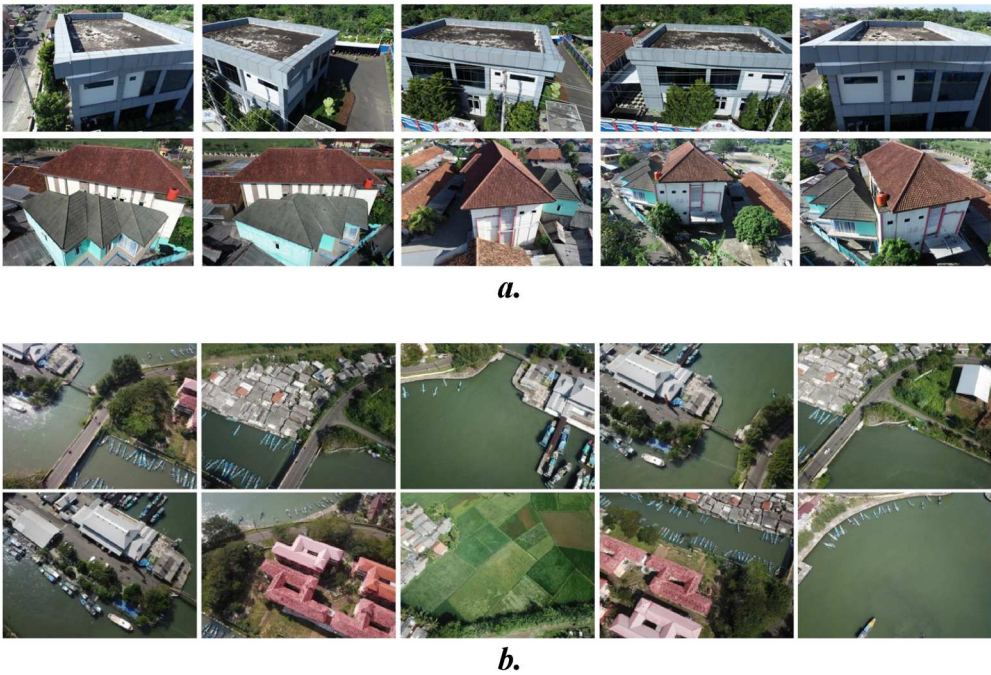


Figure 2. Aerial photos: (a) Taken by a circular mission technique. (b) Taken by a double grid technique. Source: Authors data setting.

3.2. Data collection

Drone flights were carried out from 27 May to 1 June 2023. This research used six GCP tie points installed in six locations over 30 Ha. The number and position of GCP make it possible to correct the aerial photography images obtained and increase the mapping accuracy in the photogrammetry process. The GCP pattern is drawn using white spray paint in the shape of a cross, circle and/or number. The shape of the GCP is adjusted to the autopilot tracking path in the area where the drone is flying so that the marking point can be adequately detected when the drone is in the air. The distribution pattern of those six GCP in this paper is provided in [Figure 3](#).

The DJI Phantom was the first drone to be flown to completion, after which the Mavic Pro drone was flown. Before flying the drone, the autopilot tracking data was first set using Pix4D Capture software. The data settings are shown in [Figure 4](#).

3.3. Data processing

3.3.1. From aerial photographs to 3D orthomosaic

Aerial photographs taken will be processed into a single unit in 3D orthomosaic by Agisoft Metashape software. This study used a set of multimedia computers with the following specifications: OS = Windows 64-bit, RAM = 32 GB, CPU = Intel (R) Xeon (R) CPU E3-1245 v3 @ 3.40 GHz, GPU(s) = Quadro K600. [Table 1](#) shows the processing settings in the software.

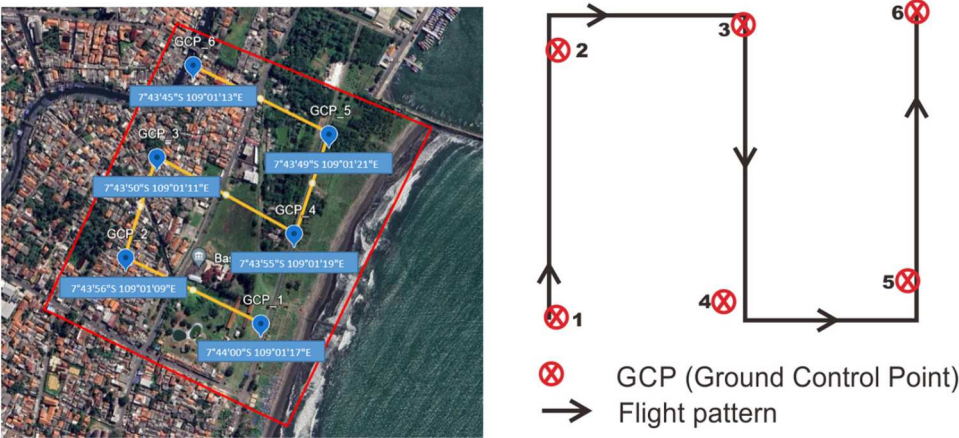


Figure 3. GCP pattern. Source: Authors data setting.

3.3.2. Tsunami magnitude prediction data

Hypothetical earthquake-tsunami based on the South Java Megathrust scenario is a maximum earthquake potential of M8.8 (Pusgen, 2017). The epicentre of the hypothetical earthquake is perpendicular to the District of Cilacap of Central Java Indonesia. It is approximately 145 km to the South-West off Cilacap of Teluk Penyau Bay.

The source parameters of the earthquake, namely the rupture of length and width as well as the other angle parameters such as strike, dip, rake based on Global Slab 1.0 (Hayes et al., 2012) and its rupture's dimension are assumed by using the scaling law (Blaser et al., 2010). The deformation model is utilised the analytical formula of the displacement fields

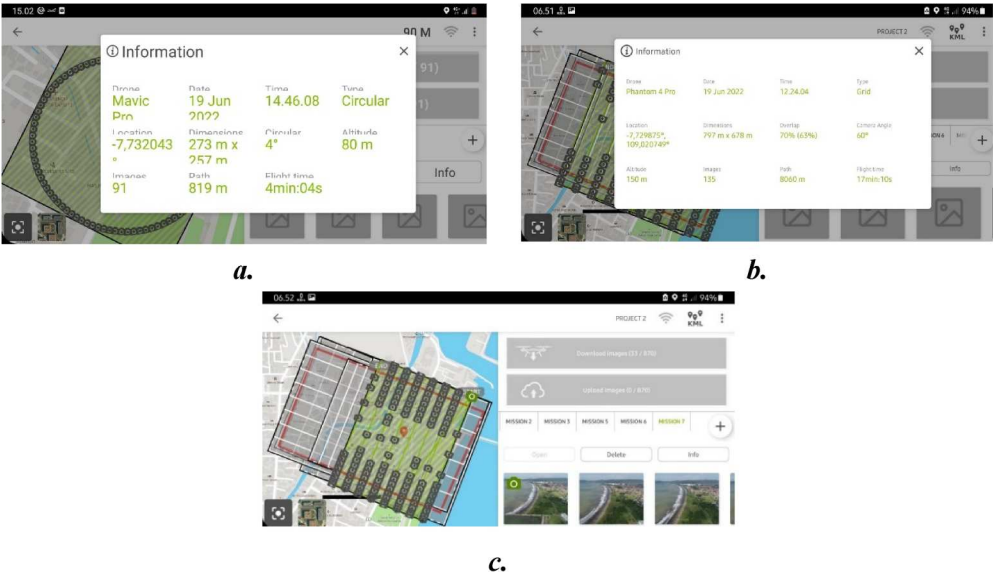


Figure 4. (a) Setting plan for the circular mission. (b) Setting plan for the double grid mission. (c) Final setting. Source: Authors data setting.

Table 1. Settings in the Agisoft metashape.

Workflow	Setting and parameters
Align Photos	Accuracy: High. Generic preselection: Yes. Reference preselection: Yes. Key point limit: 40.000. Tie point limit: 4.000.
Build Dense Cloud	Quality: High. Depth filtering: Mild. Reuse depth maps: Yes. Calculate point colours: Yes.
Build Mesh	Source data: Dense cloud. Surface type: Arbitrary (3D). Face count: High (12,000,000). Interpolation: Enabled (default). Calculate vertex colours: Yes.
Build Texture	Texture type: Diffuse map. Source data: Images. Mapping mode: Keep uv. Blending mode: Mosaic (default). Texture size/count: 4096×1 . Enable hole filing: Yes.
Build DEM	Type: Geographic. WGS 84 (EPSG: 4326). Source data: Dense cloud. Interpolation: Enabled (default). Resolution (m): 0.0584927. Total size (pix): 8088×6271 .
Build Orthomosaic	Type: Geographic. Surface: DEM. Blending mode: Mosaic (default). Enable hole filling: Yes. Pixel size ($^{\circ}$): $1.32467e-07$ (X) – $1.32224e-07$ (Y)

Source: Authors data setting.

of inclined faults (Mansinha & Smylie, 1971). The source parameter is depicted in Table 2. Meanwhile, Figure 5 shows the maximum tsunami run-up/inundation of hypothetical earthquake-tsunami M8.8 and Figure 6 shows the maximum tsunami flow depth overlaid with the ETA contour of the Hypothetical Earthquake-Tsunami M 8.8.

The geometric data for the tsunami numerical model that consist of the topographic and bathymetric data of the study area in Cilacap is a highly data resolved from GITEWS Project, which is, the data set, assimilated from field surveys as well as an LiDAR data with a resolution of less than 5 metres (Kongko, 2012). The tsunami numerical model based on the Non-Linear Shallow Water Equation (NLSWE), which is, the origin version, named TUNAMI N3 developed for TIME Project of IOC – UN (Imamura et al., 2006). It consists of 6 layers of the nested model, where the biggest area covered the megathrust of the subduction zone of South Java Indonesia, and the smallest area is the study area where those also posing the highest resolution. The results of the tsunami numerical model at three points of interest in this paper (1 coastal point, 2 of Tsunami Evacuation Shelters), such as the inundation and Estimated Time Arrival (ETA) of tsunami, are listed in Table 3.

3.3.3. Determining height and distribution of tsunami waves in the 3D orthomosaics model

Modelling the waves movement from the coastline to the final point on land uses the simple additive weighting (SAW) method. The method is the simplest and most intuitive method to handle Multiple Criteria Decision Making (MCDM) problems (Riza, 2019). In the context of disaster management, the SAW method is used to make decisions regarding resource allocation, evacuation planning, risk assessment, infrastructure resilience,

Table 2. Parameter of the hypothetical model of South Java Megathrust.

Mw	Mo	Fault	Epicenter (deg)		Mag	Depth	Focal Mechanism			Dimension		Disloc	Rigidity
		Name	Lon	Lat	(Mw)	(Km)	Strike	Dip	Slip	L(Km)	W(Km)	(m)	(Gpa)
8,41	5,16E + 21	Sub-fault-1	107,2885	−8,7439	8,4	21	293	11	90	148,00	148,00	10,00	24
8,45	5,93E + 21	Sub-fault-2	108,6616	−9,1723	8,4	24	287	11	90	148,00	148,00	10,00	27
8,45	5,88E + 21	Sub-fault-3	110,0336	−9,4581	8,4	24	281	11	90	148,00	148,00	10,00	27
8,75	1,70E + 22	Total			8,8								

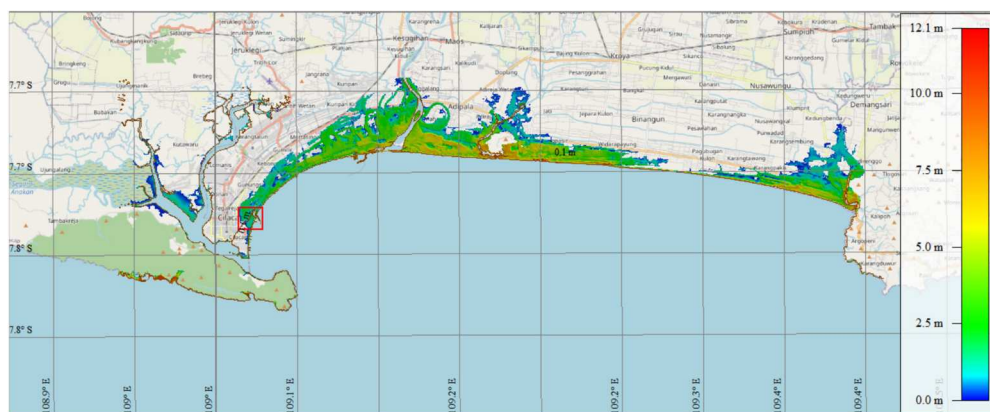


Figure 5. Maximum Tsunami Run-up/Inundation of Hypothetical Earthquake –Tsunami M8.8. Source: (Kongko, 2012).

response planning, community preparedness and distribution of humanitarian aid (Kirana et al., 2023; Nawindah, 2017; Rasadi et al., 2021). This research uses 3ds Max software to animate of wave height, inundation height, wave distribution and wave reach in 3D Orthomosaic. Figure 7 depicts the illustration of the method and Figure 8 shows the 3D Orthomosaic model from the top and circular view.

3.3.4. 3D animation tsunami modelling

This research uses Blender software to create an animation. First, we imported the 3D orthomosaic model in COLLADA (*.dae) or Wavefront OBJ (*.obj) file format. Then, we

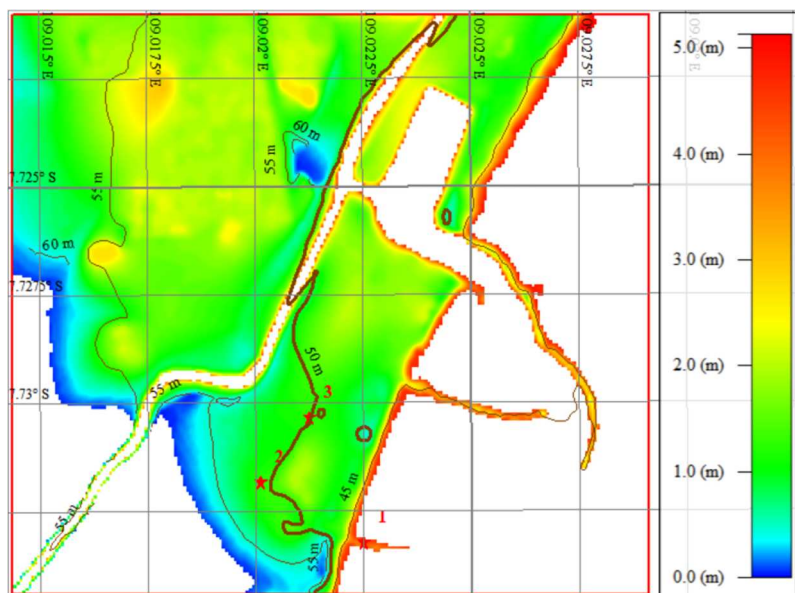


Figure 6. Maximum of Tsunami Flow depth overlaid with ETA's contour of Hypothetical Earthquake-Tsunami M 8.8.

Table 3. Tsunami numerical model at three points of interest (1 coastal point, 2 of Tsunami Evacuation Shelters).

	Point of interest	Topography (m)	Flow depth (m)	Inundation (m)	ETA (minutes)
1.	Coastal Point	0197	4355	4551	44,062
2.	TES_Dormitory_Polwan_Sanika_Satyawada	3377	1193	4570	50,529
3.	TES_PT_Sumber_Samudra_Surya_Perkasa	3323	1209	4532	50,075

designed the beach sand, sea, waves and colours. After rendering, the animated film is ready for analysis (Figure 9). The link to view the animation is <https://app.bingkaikodeku.my.id/tsunami3d/>

3.3.5. Marking evacuation routes and shelters and validation

The animated film shows the distribution of the tsunami inundation, the endpoint of the inundation, and the buildings taller than the tsunami inundation. Next, the authors marked the coordinates of those buildings and marked the paths using visual analysis techniques.

4. Results

4.1. Aerial photographs

The total of aerial photos resulted from the photogrammetry process comprises of 1099 photos, with detailed characteristics of each photo: dimensions: 5472 pixels (width) × 3648 pixels (height), horizontal and vertical resolution: 72 dpi, bit depth: 24, camera manufacturer: DJI, Camera model: FC6310, F-stop: f/5.6, exposure time: 1/500 sec, ISO speed: ISO-100, exposure bias: −0.3 steps, focal length: 9 mm, maximum aperture: 2.97, metering

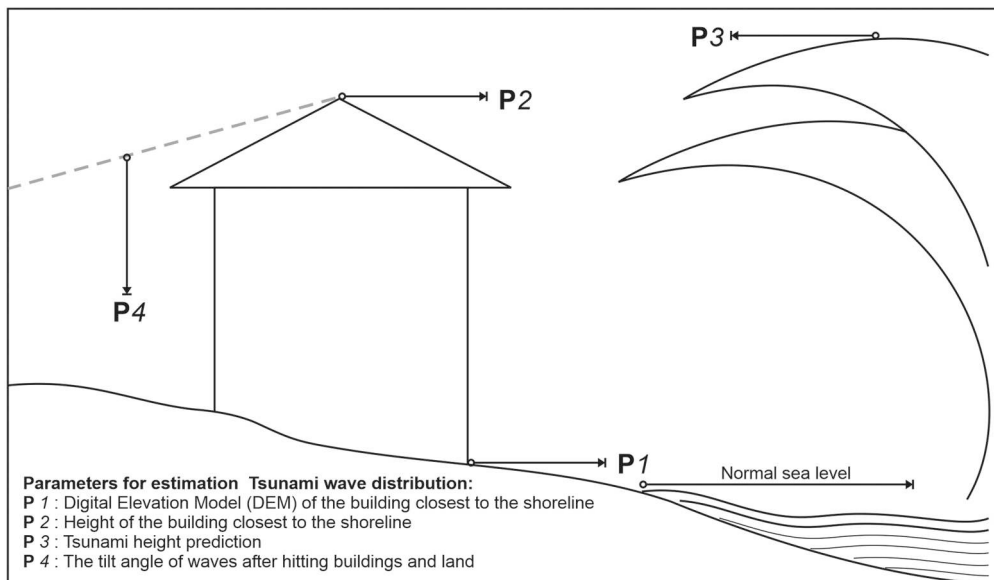
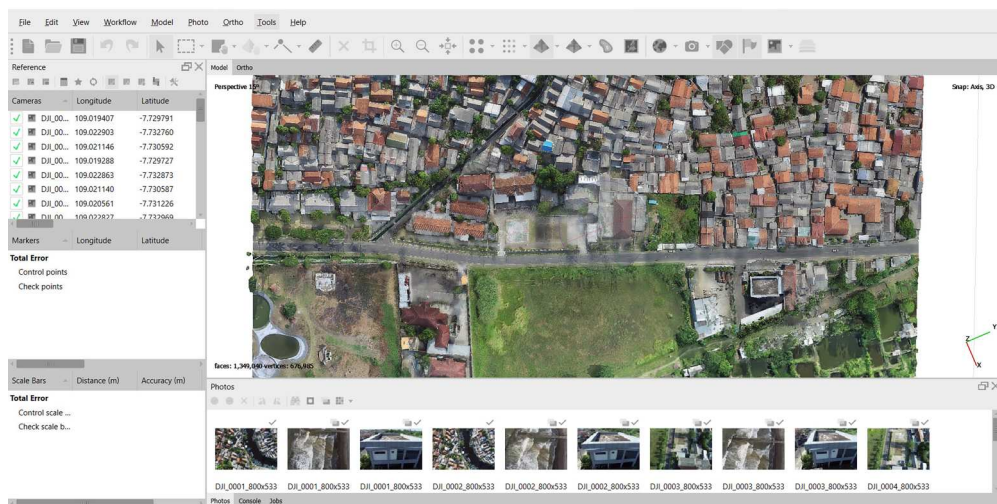
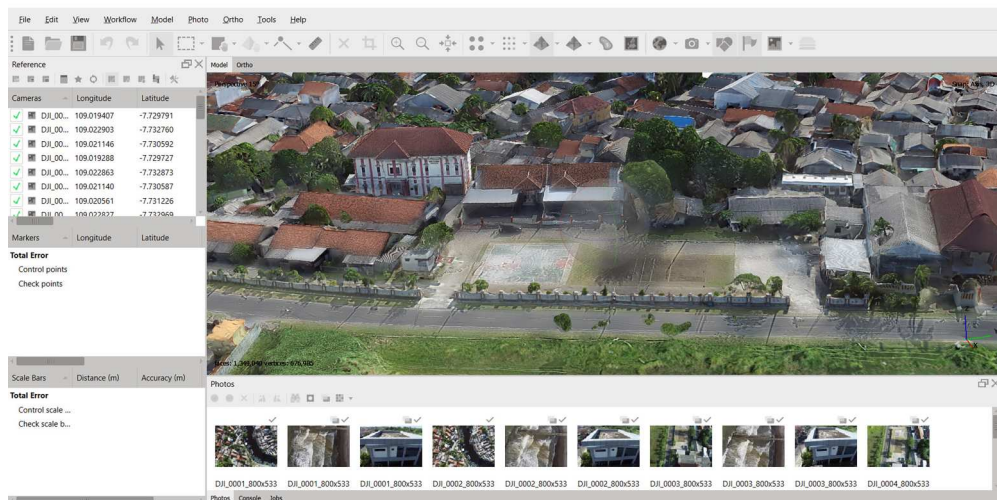


Figure 7. Illustration of the SAW method for describing tsunami magnitude in 3D orthomosaic. Source: Rezaldi et al., 2021.



a.



b.

Figure 8. (a) 3D Orthomosaic model from the top. (b) From the circular view. Source: Authors analysis.

mode: centre weighted average, subject distance: 0 mm, flash mode: no flash function, 35 mm focal length: 24, contrast: normal, light source: daylight, exposure programme: normal, saturation: normal, sharpness: normal, white balance: auto, EXIF version: 0230.

4.2. Recommended shelters and evacuation routes

The animated film shows the roofs of buildings visible as the tsunami waves spread to land. [Figure 10\(a\)](#) shows the condition of AOI before the tsunami and [Figure 10\(b\)](#) when the tsunami occurred. Meanwhile, [Figure 11\(a\)](#) shows five potential buildings used as shelters. [Figure 11\(b\)](#) shows a perspective view of one of the potential shelters.



Figure 9. (a) 3D animated tsunami modelling from top. (b) From perspective views. Source: Authors analysis.

Then, the author visited the buildings individually, measured their heights, recorded their coordinates, and checked their condition. The author used a laser telescope to measure the height of the building. Table 4 lists the recommended and non-recommended of the five buildings.

According to past event, slow earthquakes occurred in southern Java, for example the 2006 (Fritz et al., 2007) and the 1994 earthquake (Tsuji et al., 1995). As for future potential event, the slow earthquake also may be happening, and less infrastructures damage also may be expectable. However, the characteristic of slow earthquakes, as also known as tsunami earthquakes, can cause much larger tsunami (Mallick et al., 2021; Newman et al., 2011). Therefore, choosing a shelter surrounded by buildings is a good choice, because it can block the path of tsunami waves. Meanwhile, the earthquake occurred did not damage the building so it did not block the road to the shelter.

Evacuation routes are determined based on visual analysis techniques. This technique analyses images with the naked eye. This technique is often used to analyse dynamic and interactive visual interfaces (e.g. dashboards, graphs, or maps) (Ahmad et al., 2022; Caban & Gotz, 2015). The route is determined based on the closest distance to the shelter, and it is assumed that the road conditions are easy to pass, for example, the road is flat and free of potholes. Figure 12 shows the recommended shelters and the evacuation routes on foot or by vehicle.



Figure 10. (a) Conditions at AOI before the tsunami. (b) During the tsunami. Source: Authors analysis.

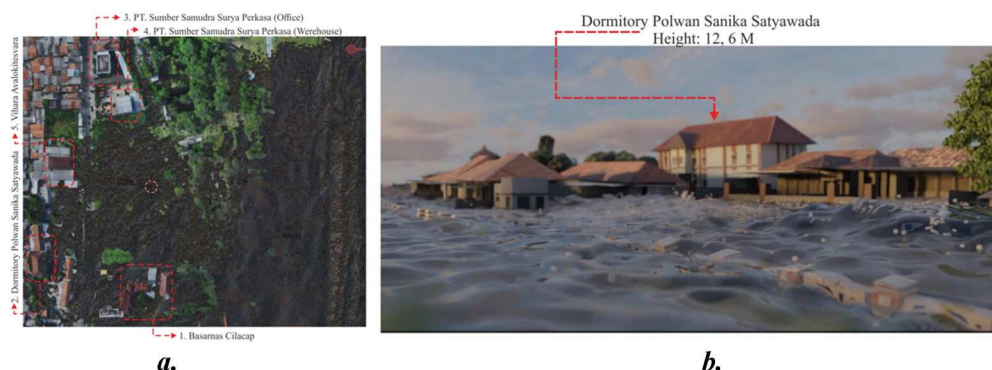


Figure 11. (a) Determination of prospective shelters from top view. (b) From perspective view. Source: Authors analysis.

Table 4. List of buildings that are taller than the tsunami inundation.

The buildings name	Coordinate points	Height	Notes
Basarnas Cilacap	Latitude: 7°26'9.34" S Longitude: 109° 0'42.65" E	6,8 m	Not recommended – even though the building height meets the requirements, the building is only 1 floor (The distance of this building from the beach is 267 m).
Dormitory Polwan Sanika Satyawada	Latitude: 7°43'54.65" S Longitude: 109° 1'12.42" E	12,6 m	Recommended – a two-story building surrounded by small buildings that can hinder the wave's speed (The distance of this building from the beach is 323 m).
PT. Sumber Samudra Surya Perkasa (Office)	Latitude: 7°43'49.22" S Longitude: 109° 1'16.49" E	9,6 m	Recommended – a two-story building surrounded by small buildings that can hinder the wave's speed (The distance of this building from the beach is 430 m).
PT. Sumber Samudra Surya Perkasa (warehouse)	Latitude: 7°26'5.98" S Longitude: 109° 0'42.60" E	7 m	Not recommended – even though the building height meets the requirements, the building is only 1 floor, and the construction seems less sturdy (The distance of this building from the beach is 280 m).
Vihara Avalokitesvara	Latitude: 7°26'5.62" S Longitude: 109° 0'41.94" E	10 m	Not recommended – even though the building height meets the requirements, the building is only 1 floor (The distance of this building from the beach is 322 m).

Source: Authors analysis.

If the earth feels like shaking and there are signs of a tsunami, such as extreme sea receding, then people on the *Teluk Penyu* beach must immediately evacuate independently. The evacuation route by walking quickly or running is 323 and 430 meters, can be taken in approximately 2 min. The route is a dirt road and is bumpy. Meanwhile, the evacuation route using a vehicle (car/motorcycle) is 675 metres, in the form of an asphalt and wide road, and can be taken in approximately 3 min, if there is no traffic jam.

5. Discussions

The use of two types of drones with different shooting angles – high and circular angle provides additional insight into orthomosaic map. The technique uses a higher level of



Figure 12. Recommended shelters and evacuation routes on foot or by vehicle. Source: Authors analysis.

overlay, thereby obtaining more image information for the software to reconstruct an object. This technique produces a more detailed 3D model. In addition, taking pictures at a circular angle clarifies information about the height of the tsunami waves on the shoreline and the height of inundation among the buildings. Meanwhile, the image taken at a high angle clarifies information on the distribution and range of waves on land. Mahendran et al. (2023) reported that circular grids play an important role because they have a better appearance compared to rectangular grids when converted into 3D models. However, there are a number of main parameters on the camera that need to be recognised by the software used so that the desired image quality is obtained, such as focal length, image size, sampling characteristics (in the case of digital source images), main point and lens distortion. Meanwhile, the use of two drones has benefits including increasing the mapping area (Avellar et al., 2015; Ju & Son, 2018), the resolution, accuracy and detail of images (Gomez & Purdie, 2016; Xiang et al., 2019; Zheng et al., 2018), reducing the data collection time (Mozaffari et al., 2018), reducing the risk of data loss – if there is a damage to a drone (Park et al., 2015) and obtaining optimal shooting angles for each drone (Wu et al., 2021).

Mapping large areas or creating maps with high accuracy requires more GCP. This research maps a relatively small AOI (30 Ha). A good orthomosaic map is obtained using only 6 GCP. Some guidelines recommend at least 5–10 GCPs for small to medium sized mapping areas (12.5–25 Km) with a maximum distance between GCP points of 2.5 Km (Yu et al., 2022). A case study proved that the use of 88 GCP on an area of 18 ha obtained the best level of calibration on aerial photo images (Zhang et al., 2022). The large number of GCP has implications for high costs and longer research time. The differences in orthomosaic maps produced when using GCP 0, 4, 8 and 16 on the same object are differences in tone on the object, visible seam lines, and objects that are distorted (Júnior et al., 2018). They recommended to test the distribution and minimum

number of GCP as a function of Ground Sampling Distance (GSD) to obtain a good map. Oniga et al. (2018) said that a minimum of 3 GCP is required for an area of less than 3 km, while it is necessary to use more GCP to increase the map accuracy. Moreover, the use of GCP distributed throughout the mapping area will help the accuracy of the map (Zhang et al., 2022).

Knowing the time of arrival of the first tsunami wave is very important for evacuation efforts. The two buildings proposed as shelters in this study are approximately 323 and 430 metres from the shoreline. Healthy adults have an average running speed of 16–24 kilometres per hour. Thus, both distances can be covered in less than 2 min. Widiyanto et al. (2019) reported that the first tsunami reached land less than 10 min after the Sulawesi earthquake in 2018. Meanwhile, Wood and Peters (2015) reported that the wave arrived within 1–2 min in a tsunami event related to an underwater landslide that was initiated by an earthquake at a subduction zone in Alaska. Regarding the incident in Portugal, models showed that the first tsunami wave arrived within 30–40 min after the earthquake (Santos et al., 2017). In the context of slow earthquakes whose vibrations are not felt by humans, such as the events in Cilacap, it is necessary to pay attention to the time it takes for tsunami waves to reach the coast after natural signs such as receding seas are visible. Based on information from several survivors, the first wave came within minutes after the sea receded (Rezaldi et al., 2023b).

The practical implication of this research is that the government must prepare buildings recommended as shelters. Apart from that, several guidelines need to be created, including a guide to evacuation routes to shelters and a map of the distribution of tsunami inundation. Meanwhile, the managerial implications include the government developing a tsunami simulation programme, training people to run from *Teluk Penyu* beach to the shelters, carrying out urban planning in the affected areas and preparing guidelines for safe or unsafe areas for building construction for the community.

6. Conclusions

This paper proposes an innovative procedure by synergizing several scientific disciplines, such as photogrammetry, disaster data and motion graphics, to produce a 3D animation for evacuation management. The aerial photos produced by the drones provide up-to-date information about an area. In addition, future tsunami magnitude data makes the analysis results relevant to future conditions. Apart from that, 3D animation makes it easier for decision-makers to make policies related to disaster evacuation management plans, such as determining evacuation routes and shelters.

Based on the research results, we recommend the following to local governments:

- The government creates and installs a sign system for tsunami evacuation routes to shelters.
- The government adjusts the number of shelters available at AOI to the number of people/tourists who need them.
- The government considers building new shelters or strengthening/renovating buildings that are not recommended in the AOI, considering the travel time for people in

the *Teluk Penyu* area to reach them and determining the shortest and barrier-free route to the new shelters.

This new research is based on visual analysis, therefore further research is recommended to test the strength of buildings recommended as shelters and measure the building capacity. Research can also be expanded to determine areas suitable for building vertical shelters. In addition, the obstacles encountered in the proposed evacuation route and the travel time required from *Teluk Penyu* beach to the shelter need to be identified.

7. Limitation

This research identifies buildings that have the potential to be used as shelters and routes to these buildings. The recommended building strength was not tested further. In addition, this study did not test evacuation routes to analyse obstacles and determine the required travel time. This test is important to find out whether a person's physical stamina in a panic condition can reach shelter if a tsunami occurs. Furthermore, the ability to carry out the evacuation on time (ET), depends on (1) the location of the tsunami safe area and its characteristics, (2) land cover, (3) topography (slope), (4) population density, (5) age distribution and gender and (6) density of important facilities (primary schools, hospitals) (Post et al., 2009). They reported that the *Teluk Penyu* coastal area had an ET value of more than 30 min, which indicates the need for special attention, good evacuation planning and a good warning dissemination structure. Therefore, a vertical shelter that can be reached in less than 2 min is important in the *Teluk Penyu* beach area, which is visited by many tourists.

Acknowledgements

The Authors pronounce many thanks to Research Centre for Data and Information Sciences – BRIN, Research Centre for Geological Disaster – BRIN, Research Centre for Hydrodynamic – BRIN, Regional Disaster Management Agency (BPBD) Cilacap – Indonesia, Programme house disaster DIPA OR KM BRIN No. SP DIPA-124.01.1.690501/2022 and for all parties who have helped carry out this research activity who cannot be mentioned one by one.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by DIPA OR KM BRIN [Grant Number: SP DIPA-124.01.1.690501/2022].

References

Affan, M., Goto, Y., & Agussabti. (2012). Tsunami evacuation simulation for disaster awareness education and mitigation planning of Banda Aceh. In *15th World Conference on Earthquake Engineering*. Vol. 24. http://www.iitk.ac.in/nicee/wcee/article/WCEE2012_3525.pdf

- Ahmad, Z., Yaacob, S., Ibrahim, R., & Fakhruddin, W. F. W. (2022). "The review for visual analytics methodology." In *HORA 2022 - 4th International Congress on Human-Computer Interaction, Optimization and Robotic Applications, Proceedings* (pp. 1–10). <https://doi.org/10.1109/HORA55278.2022.9800100>
- Amri, I., Ansariadi, Amiruddin, R., Palutturi, S., Mallongi, A., Nur, R., & Sari, N. E. (2020). The influence of disaster counseling with animation video on preparednes students in elementary school in Palu. *Medico-Legal Update*, 20(1), 1302–1306. <https://doi.org/10.37506/v20/il/2020/mlu/194482>
- Ashar, F., Amaratunga, D., & Haigh, R. (2018). Tsunami evacuation routes using network analysis: A case study in Padang. *Procedia Engineering*, 212(2017), 109–116. <https://doi.org/10.1016/j.proeng.2018.01.015>
- Avellar, G. S. C., Pereira, G. A. S., Pimenta, L. C. A., & Iscold, P. (2015). Multi-UAV routing for area coverage and remote sensing with minimum time. *Sensors*, 15(11), 27783–27803. <https://doi.org/10.3390/s151127783>
- Bertolino, M., & Tanzi, T. J. (2020). Towards 3d simulation for disaster intervention robot behaviour assessment. *Advances in Radio Science*, 18, 23–32. <https://doi.org/10.5194/ars-18-23-2020>
- Blaser, L., Krüger, F., Ohrnberger, M., & Scherbaum, F. (2010). Scaling relations of earthquake source parameter estimates with special focus on subduction environment. *Bulletin of the Seismological Society of America*, 100(6), 2914–2926. <https://doi.org/10.1785/0120100111>
- Buczkowski, A. (2022). What is the difference between orthophotos, orthophotomaps, orthomosaics and true orthophotos? *Geoawesome*, 2022. <https://geoawesomeness.com/eo-hub/what-is-the-difference-between-orthophotos-orthophotomaps-orthomosaics-and-true-orthophotos/>
- Caban, J. J., & Gotz, D. (2015). Visual analytics in healthcare – opportunities and research challenges. *Journal of the American Medical Informatics Association*, 22(2), 260–262. <https://doi.org/10.1093/jamia/ocv006>
- Danardono, D., Wibowo, A. A., Sari, D. N., Dewi, E. S. M., & Priyono, K. D. (2016). Tsunami hazard mapping based on coastal system analysis using high - Resolution unmanned aerial vehicle (UAV) imagery (Case study in Kukup Coastal Area, Gunungkidul Regency, Indonesia). *Geographia Technica*, 18(2), 51–67. <https://doi.org/10.21163/GT>
- Elhalawani, M. A., Zeidan, Z. M., & Beshr, A. A. A. (2021). Implementation of close range photogrammetry using modern non-metric digital cameras for architectural documentation. *Geodesy and Cartography*, 47(1), 45–53. <https://doi.org/10.3846/gac.2021.11269>
- Febrina, R., Evan, D. K., Afriani, L., Retno, R. M. I., Susilorini, & Fitra, H. A. (2020). The analysis of tsunami evacuation route based on geographic information system: A case study in the Coast of Lampung Bay. *IOP Conference Series: Materials Science and Engineering*, 807(012022). <https://doi.org/10.1088/1757-899X/807/1/012022>
- Fritz, H. M., Kongko, W., Moore, A., McAdoo, B., Goff, J., Harbitz, C., Uslu, B., Kalligeris, N., Suteja, D., Kalsum, K., Titov, V., Gusman, A., Latief, H., Santoso, E., Sujoko, S., Djulkarnaen, D., Sunendar, H., & Synolakis, C. (2007). Extreme runup from the 17 July 2006 Java Tsunami. *Geophysical Research Letters*, 34(12), 1–5. <https://doi.org/10.1029/2007GL029404>
- Garcia, M. V. Y., & Oliveira, H. C. (2020). The influence of ground control points configuration and camera calibration for Dtm and orthomosaic generation using imagery obtained from a low-cost Uav. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 5(1), 239–244. <https://doi.org/10.5194/isprs-annals-v-1-2020-239-2020>
- Gomez, C., & Purdie, H. (2016). UAV- based photogrammetry and geocomputing for hazards and disaster risk monitoring – A review. *Geoenvironmental Disasters*, 3(23). <https://doi.org/10.1186/s40677-016-0060-y>
- Goto, Y., Affan, M., Agussabti, Nurdin, Y., Yuliana, D. K., & Ardiansyah. (2012). Tsunami evacuation simulation for disaster education and city planning. *Journal of Disaster Research*, 7(1), 92–101. <https://doi.org/10.20965/jdr.2012.p0092>
- Goto, Y., Ogawa, Y., & Komura, T. (2010). Tsunami disaster reduction education using town watching and moving tsunami evacuation animation trial in Banda Aceh. *Journal of Earthquake and Tsunami*, 4(2), 115–126. <https://doi.org/10.1142/S1793431110000728>
- Gurjar, S. P. S., Jain, K., & Mandla, V. R. (2013). Virtual 3D campus modeling by using close range photogrammetry. *American Journal of Civil Engineering and Architecture*, 1(6), 200–205. <https://doi.org/10.12691/ajcea-1-6-9>

- Hayes, G. P., Wald, D. J., & Johnson, R. L. (2012). Slab1.0: A three-dimensional model of global subduction zone geometries. *Journal of Geophysical Research: Solid Earth*, 117(B1), B01302. <https://doi.org/10.1029/2011JB008524>
- Hébert, H., Burg, P. E., Binet, R., Lavigne, F., Allgeyer, S., & Schindelé, F. (2012). The 2006 July 17 Java (Indonesia) tsunami from satellite imagery and numerical modelling: A single or complex source? *Geophysical Journal International*, 191(3), 1255–1271. <https://doi.org/10.1111/j.1365-246X.2012.05666.x>
- Hidayat, R., Wibowo, M., & Sartimbul, A. (2021). Simulation of oil spill pollution due to tsunami in cilacap, central java, Indonesia. *International Journal on Advanced Science, Engineering and Information Technology*, 11(3), 898–906. <https://doi.org/10.18517/ijaseit.11.3.14264>
- Idrus, F. N., Zuhriyah, L., & Suharsono, T. (2019). Comparison of simulation method and animation video on knowledge related to preparedness of elementary school students in ternate, Indonesia. *Indian Journal of Public Health Research and Development*, 10(8), 2037–2041. <https://doi.org/10.5958/0976-5506.2019.02154.5>
- Imamura, F., Yalciner, A. C., & Ozyurt, G. (2006). *Tsunami modelling manual: (TUNAMI model)*. Tohoku University.
- Indira, I., Dwi, M., & Manessa, M. (2023). Generating evacuation route for tsunami evacuation based on megathrust scenario hazard model in Palabuhanratu Village, Sukabumi, West Java. *International Journal of Disaster Management*, 6(1), 35–48. <https://doi.org/10.24815/ijdm.v6i1.31148>
- Jati, B. A. E. K., Al Akbar, M. F., Wahyuni, T., Khasanah, E. U., Paramanandi, A. R. G., Sutiono, H., E., C., P., Setyaningsih, D., P., Widyatmanti, W., Wibowo, T., W. (2023). Analysis of tsunami evacuation route planning in Kulon Progo Regency. *International Journal of Remote Sensing and Earth Sciences*, 20(1), 16–25.
- Ju, C., & Son, H. I. (2018). Multiple UAV systems for agricultural applications: Control, implementation, and evaluation. *Electronics*, 7(162), 1–19. <https://doi.org/10.3390/electronics7090162>
- Júnior, L. R. A., Ferreira, M. E., Côrtes, J. B. R., & de Castro Jorge, L. A. (2018). High accuracy mapping with cartographic assessment for a fixed-wing remotely piloted aircraft system. *Journal of Applied Remote Sensing*, 12(1), 1–21. <https://doi.org/10.1117/1.JRS.12.014003>
- Kawai, J., Mitsuhashi, H., & Shishibori, M. (2016). Management (ICT-DM) - Tsunami evacuation drill system using motion hazard map and smart devices. In *IEEE 2016 3rd International Conference on Information and Communication Technologies for Disaster Management (ICT-DM) - Vienna, Austria*.
- Kerta, W. I. D. K., Abdul, K. F., & Huda, F. M. (2021). The preparedness of communities around cilacap industrial companies in facing tsunami treats. *Disaster Advances*, 14(9), 49–56. <https://doi.org/10.25303/149da4956>
- Kim, T. H., Cova, T. J., & Brunelle, A. (2006). Exploratory map animation for post-event analysis of wildfire protective action recommendations. *Natural Hazards Review*, 7(1), 1–11. [https://doi.org/10.1061/\(ASCE\)1527-6988\(2006\)7:1\(1\)](https://doi.org/10.1061/(ASCE)1527-6988(2006)7:1(1))
- Kirana, A. P., Vista, C. B., Eliyen, K., Wijayaningrum, V. N., Firdausi, A. T., & Susanti . (2023). Geospatial analysis and mapping of natural disasters in Situbondo, Indonesia: A Simple Additive Weighting (SAW) Approach. In *8th International Conference on Electrical, Electronics and Information Engineering (ICEEIE)*. <https://doi.org/10.1109/ICEEIE59078.2023.10334889>
- Kongko, W. (2012). *South Java tsunami model using highly resolved data and probable tsunamigenic sources*. University of Hannover.
- Laksono, F. A. T., Widagdo, A., Aditama, M. R., Fauzan, M. R., & Kovács, J. (2022). Tsunami hazard zone and multiple scenarios of tsunami evacuation route at Jetis Beach, Cilacap Regency, Indonesia. *Sustainability (Switzerland)*, 14(5), 1–19. <https://doi.org/10.3390/su14052726>
- Mahendran, S., Eswar, K., Praveen Kumar, R., Dineshwaran, S., Vinyojita, M., Asokan, R., & Seralathan, S. (2023). Experimental investigations to enhance precision and accuracy in UAV photogrammetry of 3D model. In *IEEE International Conference on Electrical, Electronics, Communication and Computers, ELEXCOM 2023*. <https://doi.org/10.1109/ELEXCOM58812.2023.10370555>
- Mallick, R., Meltzner, A. J., Tsang, L. L. H., Lindsey, E. O., Feng, L., & Hill, E. M. (2021). Long-lived shallow slow-slip events on the Sunda Megathrust. *Nature Geoscience*, 14(5), 327–333. <https://doi.org/10.1038/s41561-021-00727-y>

- Mansinha, L., & Smylie, D. (1971). The displacement field of inclined faults. *Bulletin of the Seismological Society of America*, 61(5), 1433–1440. <https://doi.org/10.1785/BSSA0610051433>.
- Marfai, M. A., Fatchurohman, H., & Cahyadi, A. (2019a). An evaluation of tsunami hazard modeling in Gunungkidul Coastal Area using UAV photogrammetry and GIS. Case study: Drini Coastal Area. In *E3S Web of Conferences*. Vol. 125. <https://doi.org/10.1051/e3sconf/201912509005>
- Marfai, M. A., Khakim, N., Fatchurohman, H., & Salma, A. D. (2021). Planning tsunami vertical evacuation routes using high-resolution UAV digital elevation model: Case study in Drini Coastal Area, Java, Indonesia. *Arabian Journal of Geosciences*, 14(19), 2028. <https://doi.org/10.1007/s12517-021-08357-9>
- Marfai, M. A., Sunarto, Khakim, N., Cahyadi, A., Rosaji, F. S. C., Fatchurohman, H., & Wibowo, Y. A. (2018). Topographic data acquisition in tsunami-prone coastal area using Unmanned Aerial Vehicle (UAV). *IOP Conference Series: Earth and Environmental Science*, 148(April), 012004. <https://doi.org/10.1088/1755-1315/148/1/012004>
- Marfai, M. A., Sunarto, S., Khakim, N., Fatchurohman, H., Cahyadi, A., Wibowo, Y. A., & Rosaji, F. S. C. (2019b). Tsunami hazard mapping and loss estimation using geographic information system in Drini Beach, Gunungkidul Coastal Area, Yogyakarta, Indonesia. In *E3S Web of Conferences* (p. 76). <https://doi.org/10.1051/e3sconf/20197603010>
- Marras, S., & Mandli, K. T. (2021). Modeling and simulation of tsunami impact: A short review of recent advances and future challenges. *Geosciences (Switzerland)*, 11(1), 1–19.
- Menner, A. W. (1995). Introduction to modeling and simulation techniques. *JOHNS HOPKINS APL TECHNICAL DIGEST*, 16(1), 6–17.
- Mozaffari, M., Saad, W., Bennis, M., & Debbah, M. (2018). Drone-based antenna array for service time minimization in wireless networks. In *Proc. IEEE Int. Conf. Commun. (ICC)* (pp. 1–6). <https://doi.org/10.1109/ICC.2018.8422374>
- Nawindah, N. (2017). Simple additive weighting (SAW) mathematics method for warehouse disaster location selection in Central Jakarta, Indonesia. *International Journal of Pure and Applied Mathematics*, 117(15), 795–803.
- Newman, A. V., Hayes, G., Wei, Y., & Convers, J. (2011). The 25 October 2010 Mentawai tsunami earthquake, from real-time discriminants, finite-fault rupture, and tsunami excitation. *Geophysical Research Letters*, 38(5), 5–11. <https://doi.org/10.1029/2010GL046498>
- Nogueira, F. d. C., Roberto, L., Korting, T. S., & Shiguemori, E. H. (2017). Accuracy analysis of orthomosaic and DSM produced from sensor aboard UAV. In *Conference: Brazilian Symposium on Remote Sensing*. Santos: Galoa.
- Nurwatik, N., & Setiawan, D. (2023). Using GIS to determine tsunami shelter location in South Coast Pangandaran District, Pangandaran Regency, Indonesia. *IOP Conference Series: Earth and Environmental Science*, 1127(012048). <https://doi.org/10.1088/1755-1315/1127/1/012048>
- Oniga, V.-E., Breaban, A.-I., & Statescu, F. (2018). Determining the optimum number of ground control points for obtaining high precision results based on UAS images. *Proceedings*, 2(352), 2–11. <https://doi.org/10.3390/ecrs-2-05165>
- Pajorová, E., Hluchý, L., Halada, L., & Slížik, P. (2008). 3D visualization tool for virtual models of natural disasters. In *Environmental Economics and Investment Assessment II*, no. January 2007 (pp. 105–14).
- Park, C., Cho, N., Lee, K., & Kim, Y. (2015). Formation flight of multiple UAVs via onboard sensor information sharing. *Sensors (Switzerland)*, 15(7), 17397–17419. <https://doi.org/10.3390/s150717397>
- Post, J., Wegscheider, S., Mück, M., Zosseder, K., Kiefl, R., Steinmetz, T., & Strunz, G. (2009). Assessment of human immediate response capability related to tsunami threats in Indonesia at a Sub-National Scale. *Natural Hazards and Earth System Science*, 9(4), 1075–1086. <https://doi.org/10.5194/nhess-9-1075-2009>
- Prasetyadi, A., Rezaldi, M. Y., Yoganingrum, A., Hanifa, N. R., & Kongko, W. (2022). UAV-Photogrammetry specification for generating 3D-orthomosaic in case of 3D animation modeling. In *ACM International Conference Proceeding Series*. <https://doi.org/10.1145/3575882.3575933>
- PuSGeN. (2017). Peta Sumber dan Bahaya Gempa. Indonesia Tahun 2017. Pusat Penelitian dan Pengembangan Perumahan dan Permukiman Badan Penelitian dan Pengembangan Kementerian Pekerjaan Umum dan Perumahan Rakyat: Jakarta.

- Rasadi, A., Hidayat, B., & Ophiandri, T. (2021). Decision support system in determining the priority of disaster mitigation infrastructure development in villages level using the simple additive weight (SAW) method. *IOP Conference Series: Earth and Environmental Science*, 708(012065). <https://doi.org/10.1088/1755-1315/708/1/012065>
- Rezaldi, M. Y., Kadir, R. A., Ijab, M. T., & Apip. (2023a). Aesthetics 3D geovisualization for flood disaster based on the XYZ coordinate. *Journal of Applied Research and Technology*, 21(4), 688–706. <https://doi.org/10.22201/icat.24486736e.2023.21.4.1706>
- Rezaldi, M. Y., Yoganingrum, A., Hanifa, N. R., Kaneda, Y., Kushadiani, S. K., Prasetyadi, A., Nugroho, B., & Riyanto, A. M. (2021). Unmanned Aerial Vehicle (Uav) and photogrammetric technic for 3d tsunami safety modeling in Cilacap, Indonesia. *Applied Sciences (Switzerland)*, 11(23), 11310. <https://doi.org/10.3390/app112311310>
- Rezaldi, M. Y., Yoganingrum, A., Hanifa, N. R., Prasetyadi, A., Kongko, W., & Kaneda, Y. (2023b). The natural warning signs of tsunami earthquake in Indonesia: Case of the 2006 Cilacap event. *Environmental Hazards*, 22(5), 456–474. <https://doi.org/10.1080/17477891.2023.2190871>
- Riza, M. W. A. (2019). Design and analysis of decision support information system to determine the priority of selecting natural disaster barrack policies with the fuzzy simple additive weighting method. *JAREE (Journal on Advanced Research in Electrical Engineering)*, 3(2), 0–5. <https://doi.org/10.12962/j25796216.v3.i2.88>
- Salmon, J. D. (1984). Vertical evacuation in hurricanes: An urgent policy problem for coastal managers. *Coastal Zone Management Journal*, 12(2–3), 287–300. <https://doi.org/10.1080/08920758409361966>
- Santos, A., Fonseca, N., Queirós, M., Zêzere, J. L., & Bucho, J. L. (2017). Implementation of tsunami evacuation maps at Setubal Municipality, Portugal. *Geosciences (Switzerland)*, 7(4), 1–19. <https://doi.org/10.3390/geosciences7040116>
- Shuto, N. (2019). Tsunami hazard mitigation. *Proceedings of the Japan Academy Series B: Physical and Biological Sciences*, 95(4), 151–164. <https://doi.org/10.2183/pjab.95.012>
- Sun, M.-C., Sakai, K., Chen, A. Y., & Hsu, Y.-T. (2022). Location problems of vertical evacuation structures for dam-failure floods: Considering shelter-in-place and horizontal evacuation. *International Journal of Disaster Risk Reduction*, 77(July), 103044. <https://doi.org/10.1016/j.ijdr.2022.103044>
- Traya, R. V., Verona, R. M. P., Malatbalat, L. A. A., Nuevas, L. K., Obediencia, D. C., Velarde, M. W. C., & Daylo, R. M. (2022). Android mobile application: Tsunami alert system with an escape route for evacuation in municipal disaster risk reduction and management office. In *2022 5th International Conference on Vocational Education and Electrical Engineering: The Future of Electrical Engineering, Informatics, and Educational Technology Through the Freedom of Study in the Post-Pandemic Era, ICVEE 2022 – Proceeding* (pp. 30–35). <https://doi.org/10.1109/ICVEE57061.2022.9930386>
- Tsuji, Y., Imamura, F., Matsutomi, H., Synolakis, C. E., Nanang, P. T., Jumadi, Harada, S., Han, S. S., Arai, K., & Cook, B. (1995). Field survey of the East Java earthquake and tsunami of June 3, 1994. *pure and Applied Geophysics*, 144(3–4), 839–854. <https://doi.org/10.1007/BF00874397>
- Usman, F., Murakami, K., Wicaksono, A. D., & Setiawan, E. (2017). Application of agent-based model simulation for tsunami evacuation in Pacitan, Indonesia. *MATEC Web of Conferences*, 97, (01064). <https://doi.org/10.1051/mateconf/20179701064>
- Vilariño, L. D., Gonzalez, H., Martínez-Sánchez, J., & Bueno, M. (2016). Determining the limits of unmanned aerial photogrammetry for the evaluation of road runoff. *Measurement*, 85, 132–141. <https://doi.org/10.1016/j.measurement.2016.02.030>
- Widiyanto, W., Santoso, P. B., Hsiao, S. C., & Imananta, R. T. (2019). Post-event field survey of 28 September 2018 Sulawesi earthquake and tsunami. *Natural Hazards and Earth System Sciences*, 19(12), 2781–2794. <https://doi.org/10.5194/nhess-19-2781-2019>
- Widiyantoro, S., Gunawan, E., Muhari, A., Rawlinson, N., Mori, J., & Hanifa, N. R. (2020). Implications for megathrust earthquakes and tsunami from Seismic Gaps South of Java Indonesia. *Scientific Reports*, 10(15274), 1–11. <https://doi.org/10.1038/s41598-020-72142-z>
- Wolf, P. R., Dewitt, B. A., & Wilkinson, B. E. (2014). *Elements of photogrammetry with application in GIS*. *Elements of photogrammetry with application in GIS* (4th ed.). Mc Graw Hill Education. <https://www.accessengineeringlibrary.com/binary/mheaeworks/beb61e95f877ce90/6d34644067d3fb5bcddec1baeb03abf1474b14cb9f44007d35502759b590b2efa/book-summary.pdf>

- Wood, N., Jones, J., Schelling, J., & Schmidtlein, M. (2014). Tsunami vertical-evacuation planning in the U.S. Pacific northwest as a geospatial, multi-criteria decision problem. *International Journal of Disaster Risk Reduction*, 9(September), 68–83. <https://doi.org/10.1016/j.ijdrr.2014.04.009>
- Wood, N. J., & Peters, J. (2015). Variations in population vulnerability to tectonic and landslide-related tsunami hazards in Alaska. *Natural Hazards*, 75(2), 1811–1831. <https://doi.org/10.1007/s11069-014-1399-6>
- Wu, J., Yu, Y., Ma, J., Wu, J., Han, G., Shi, J., & Gao, L. (2021). Autonomous cooperative flocking for heterogeneous unmanned aerial vehicle group. *IEEE Transactions on Vehicular Technology*, 70(12), 12477–12490. <https://doi.org/10.1109/TVT.2021.3124898>
- Xiang, T.-Z., Xia, G.-S., & Zhang, L. (2019). Mini-Unmanned aerial vehicle-based remote sensing: Techniques, applications, and prospects. *IEEE Geoscience and Remote Sensing Magazine*, 7(3), 29–63. <https://doi.org/10.1109/MGRS.2019.2918840>
- Yoganingrum, A. (2018). Selecting between animation and printed comic for transferring knowledge to rural schoolchildren. In *Proceedings of the European Conference on Knowledge Management, ECKM*. Vol. 2.
- Yu, X., Ge, Y., Zhao, T., Zhang, G., Liu, Y., & Yu, C. (2022). The dynamic displacement monitoring method using unmanned aerial vehicles based on digital close-range photogrammetry. *Hindawi. Mathematical Problems in Engineering*, 2022, (1949213). <https://doi.org/10.1155/2022/1949213>.
- Zhang, K., Okazawa, H., Hayashi, K., Hayashi, T., Fiwa, L., & Maskey, S. (2022). Optimization of ground control point distribution for unmanned aerial vehicle photogrammetry for inaccessible fields. *Sustainability (Switzerland)*, 14(15), 9505. <https://doi.org/10.3390/su14159505>.
- Zheng, X., Wang, F., & Li, Z. (2018). A multi-UAV cooperative route planning methodology for 3D fine-resolution building model reconstruction. *ISPRS Journal of Photogrammetry and Remote Sensing*, 146(November), 483–494. <https://doi.org/10.1016/j.isprsjprs.2018.11.004>